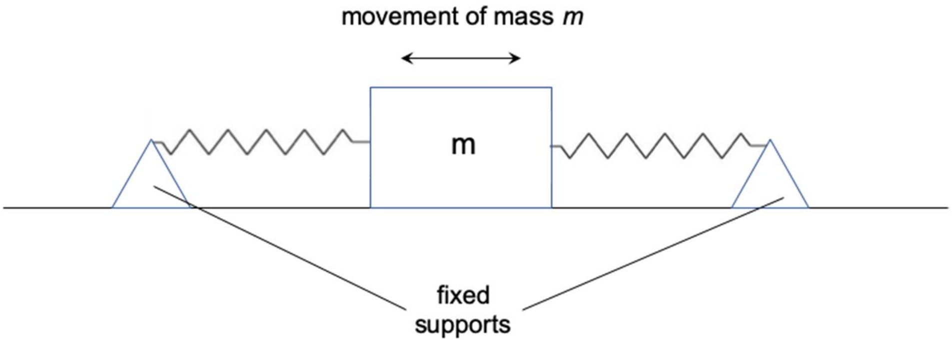
| | | |
|---|---|---|
| **16** | A mass *m* on a smooth horizontal table is attached by two light springs to two fixed supports as shown below. The mass executes simple harmonic motion of amplitude *a* and period *T*.   The total energy of oscillation is | |
| | **A** | $2\pi ma^2/T^2$ |
| | **B** | $2\pi m^2a^2/T$ |
| | **C** | $\pi^2ma^2/T^2$ |
| | **D** | $2\pi^2ma^2/T^2$ |
| | **Answer: D**<br><br>The total energy of oscillation is equal to the **maximum** KE of the object in the oscillation.<br><br>$KE_{max} = \frac{1}{2}mv_0^2$<br>$= \frac{1}{2}m(x_0\omega)^2$<br>$= \frac{1}{2}ma^2\left(\frac{2\pi}{T}\right)^2$<br>$= \frac{1}{2}ma^2\left(\frac{4\pi^2}{T^2}\right)$<br>$= \frac{2ma^2\pi^2}{T^2}$ | |